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Secondary-to-tertiary comparison through the lens of ways of doing mathematics in relation to functions: a study in collaboration with teachers

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Abstract This article addresses the issue of transition from secondary to post-secondary education through collaborative research with teachers from both levels. It takes into account implicit elements in this transition. Research on the transition in mathematics education tends to focus more on the tertiary level, studying difficulties encountered by students or comparing curricula, tasks, and textbooks highlighting disruptions or cultural shifts. However, according to Hall (1959), and Artigue (2004) in a mathematical context, it is the implicit "ways of doing" that lead to the greatest cultural differences. In this article, attention is paid to the particular ways in which teachers' ways of doing mathematics impact the transition. Ethnomethodology (Garfinkel, 1967) and its central concept of ethnomethods are used as the theoretical framework to conceptualize these ways of doing mathematics. A *breaching situation* presented to a group of secondary- and tertiary-level teachers engaged in collaborative research (Desgagné et al., 2001) brings to light familiar ways of doing mathematics. Our analysis focuses specifically on a comparison of the ways of doing mathematics related to functions at each level.

Keywords Secondary-to-tertiary transition · Mathematics culture · Ways of doing mathematics · Ethnomethodology · Mathematical ethnomethod · Functions

1 Introduction

Researchers in numerous countries indicate that the transition from secondary to tertiary education is difficult for students, who do not always understand the new *rules of the game* (e.g., Coulon, 1993; Tinto, 1993). This is especially the case in mathematics, where the

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transition constitutes a major obstacle to student success (Artigue, 2004; Corriveau, 2007; Praslon, 2000, Vandebrouck, 2011b). An analysis of secondary-to-tertiary transition research (see meta-analysis by Gueudet, 2008) shows that researchers have approached the issue from the perspectives of advanced mathematics (Robert, 1998; Tall, 1991); student difficulties with mathematical content, such as linear algebra (De Vleeschouwer & Gueudet, 2010; Dorier, 1997), functions (Vandebrouck, 2011b), and systems of equations (Stadler, 2010); difficulties with logic, proofs and formalism (Chellougui, 2004; Corriveau, 2007; Moore, 1994; Sawagedo, 2014); and knowledge organization[1] (Lithner, 2000; Sierpinska, 2000). Other studies have focused on comparing the tasks that students are expected to complete at both levels. Bosch, Fonseca, and Gascon (2004), Gueudet (2004), Praslon (2000), and Winsløw (2007) have attempted to characterize mathematical praxeologies (Bosch & Chevallard, 1999) of tasks in textbooks of each level. The authors identified differences in mathematics praxeologies at the secondary and tertiary levels for seemingly common content. Artigue (2004) refers to Hall's tripartite theory (1959) on culture to describe what she calls *mathematics culture* at the secondary level, based on an analysis of the French curriculum. Starting at the secondary level, she observes a change in mathematics culture between secondary- and tertiary-level education.

According to Hall (1959), implicit rules and ways of doing lead to the greatest cross-cultural differences. So as to contribute to the understanding of transition phenomena, we investigate these implicit rules and teachers' ways of doing at both levels. In Section 2 of this article, we present the cultural perspective adopted to address transitional issues and the ethnomethodological perspective to conceptualize the said *ways of doing mathematics* (WDMs). Section 3 presents the methodology of the collaborative research carried out. Section 4 is devoted to the analysis of the ways of doing mathematics actualized by teachers of both levels, with a specific focus on function. Then in Section 5, the results obtained are interpreted and discussed. But firstly, we will discuss how teachers' ways of doing mathematics can impact the transition.

2 Teachers and the secondary-to-tertiary transition

Although institutional elements such as coordination between curriculum and textbooks immediately come to mind when dealing with transitional issues, the subject transcends questions regarding the educational institution or development of mathematical content. Teachers can certainly aid in the understanding of transitional issues. As Roditi (2013) points out, teachers are doing much more than simply performing tasks and providing official instructions:

> Teachers practice […] their profession by referring to collectively constructed behaviors, [which] confer a social dimension onto their practices and lead them to respond to the requirements of the workplace, primarily the educational institution (management, teaching team, other professionals, etc.) and classes under their authority (level, number of students, more or less heterogeneous academic or social composition, etc.).[2]

[1] Comparison between cognitive resources used in the resolution of problems by mathematicians and students.
[2] Translated from French.

The collectively constructed practices of teachers, within the context of their teaching, i.e., mathematics education at a given level, must be taken into account when considering transitional issues. In other words, ways of doing mathematics are built through a dialectical relationship with the institution, but they also structure it (Lave, 1988). Institutions are thus seen as living organizations or communities of practice, rather than as structured, formal and determined systems isolated from the activities of those who constitute them (Cobb, McClain, de Silvia Lamberg, & Dean, 2003). This aspect appears important to creating a better understanding of the changes students face during the transition.

When interviewed about students' difficulties, tertiary teachers focus much more on difficulties in terms of ways of doing (such as ways of using symbolism, solving a problem, proving, etc.) rather than in terms of mathematical content (Corriveau, 2007). Drouhard (2006), referring to the notion of the "rules of the mathematics game," advances the hypothesis that as far as transitions are concerned, changing the rules constitutes a much larger obstacle than simply expanding and developing the study of mathematical objects. For Drouhard, these rules are implicit in instruction. Teachers therefore become key actors in studying the transition, since their use of the rules and their WDMs are often implicit. WDMs belong to the informal level of the mathematics culture, specifically non-explicit ways of acting, thinking, and reasoning resulting from experience (Artigue, 2004) and context. In this sense, the ways in which teachers do mathematics with their students and have them to do mathematics are undoubtedly different and important to investigate. Indeed, didactic and pedagogical training of secondary school teachers and disciplinary training of college teachers suggest a paradigm shift in the transition from one level to another, with new ways of approaching and doing mathematics (Durand-Guerrier, 2003). The objective of this study is more specifically to compare the secondary and tertiary levels with teachers through their ways of doing mathematics. How can teachers' WDMs be characterized at each level? How are their WDMs similar or different?

3 Theoretical basis

Our study is primarily based on Hall's theory of culture (1959) as a theoretical basis for approaching the transition as a cultural change (Artigue, 2004). According to Hall, culture is comprised of three interconnected dimensions: *formal*, *informal*, and *technical*. These make up a cohesive, structured whole.

3.1 The three dimensions of mathematics culture

The *formal dimensions* of culture seem, for people who live in this culture, natural and evident. Learning activities of this dimension are, says Hall (1959), done through injunctions and admonitions: e.g., "We do not do that, it is forbidden!" In mathematics culture, the formal level corresponds to mathematical beliefs, convictions and assumed facts; it represents a key part of mathematics culture since it defines what is acceptable and what is not without the need for justification. Mathematics teachers, in a teaching context, do mathematics relying on obvious beliefs and values without ever really questioning them. The *informal dimension* of culture refers to what

surrounds actions and is of an implicit nature. In a mathematics context, this means WDMs delineated by implicit rules and, more specifically, to ways of implementation in the teaching of mathematics without being explicit: how to work in/with a certain representation, how to introduce and use symbolism, etc. This requires investigation with teachers into their ways of doing. The *technical dimension* is fully explicit. It refers to a system of references submitted to explicit and organized logical analysis and to knowledge that is sufficiently clear to be made explicit in writing. In a mathematics culture, it refers to a system of explicit explanations, organized and institutionalized in the form of curricula and textbooks.

Hall's three dimensions of culture allow WDMs to be approached as a connected whole: the ways of doing, beliefs, assumed facts, and explicit explanations that constitute the mathematics culture at each level. However, the informal dimension, which expresses mainly implicit ways of doing, is not easily captured within this theory. Our research draws on a second theoretical foundation, *ethnomethodology* (Garfinkel, 1967), in order to examine in more detail WDMs which teachers establish and enact daily.

3.2 WDMs and ethnomethodology

Ethnomethodology[3] is a sociological framework that aims to study the social world as it continuously develops. Analysis of this world is accomplished through *ethnomethods*, which are the methods that actors use to complete their daily activities. In Garfinkel's words:

> Ethnomethodological studies analyze everyday activities as members' methods for making those same activities visibly-rational-and-reportable-for-all-practical-purposes, i.e., accountable, as organizations of commonplace everyday activities (Garfinkel, 1967, p. vii).

Ethnomethodology therefore serves as a conceptual basis for the study of common actions carried out in daily life (Coulon, 1993), e.g., sharing the road as pedestrians, cyclists or car drivers, or professional activities such as teaching mathematics at a particular level. The central concept of ethnomethods allows for closer examination of how WDMs at each level are constituted through the daily activities of teachers (see Corriveau, 2013). Ethnomethods are defined through a variety of interrelated concepts: accountability, indexicality, circumstances, interpretative procedures, and membership.

Accountability in ethnomethodology refers to the fact that an action creates an observable meaning. Members, in an ethnomethodological sense, who share a context, a language and actions, can recognize the sense of the actions of other members. The action is linked to rationality and based on the *interpretative procedures* (Cicourel, 1970 in Coulon, 1993) of actors who investigate the world and recognize the specific *circumstances* that generate particular actions (circumstances which somewhat define WDMs). Garfinkel considers interpretative procedures to be reflexive instructions which actors give each other in order to be able to understand one another and make decisions in response to their respective actions (Coulon, 1993). This perspective

[3] Though its name suggests otherwise, ethnomethodology is not a methodological approach but rather a theory of social phenomena (ethnomethodology is the study of ethnomethods).

refers back to the idea of exchanges between members, in an ethnomethodological sense, where communication plays an important role (Coulon, 1993). Ethnomethodology is not concerned with the individual actor, but rather with actors who share ethnomethods and are engaged in exchanges.

In ethnomethodology, *membership* is strongly linked to a familiarity with which other members identify. Teachers at both levels share the same context (curriculum, textbooks, didactic, and pedagogical training for secondary teachers and disciplinary training for tertiary teachers) and establish, in ways that are consistent with this context, methods for working with the mathematical content and procedures they identify with as members of that level. Their ways of doing mathematics are linked to the circumstances of their context. Actions are therefore indexed to situations and circumstances. *Indexicality* refers to the incompleteness of interactions, which can only derive their full meaning from the context of their occurrence (Coulon, 1987). Just as interpretive procedures allow actors to process actions, actors' interpretative procedures allow them to complete information from indexical expressions.

From the concept of ethnomethod, I have developed the concept of *mathematical ethnomethods*, which are WDMs mobilized by teachers in their daily working lives. The mathematical ethnomethods include action: WDMs *shared* by teachers of a specific level (membership), indexed to circumstances of this action. They also include actor considerations: the teachers are assumed to be knowledgeable, meaning that *doing mathematics* requires an interpretation from the teacher to give meaning to what they do. Table 1 introduces the concept of mathematical ethnomethods (Corriveau, 2013).

These different theoretical categories allow the transitional context to be taken into account and highlight how WDMs are established as activities structured through different interrelated concepts.

4 Methods

Hall notes that "[o]ne of the most effective ways to learn about [one's culture] is by taking seriously the cultures of others. It forces you to pay attention to those details of life which differentiate them from you" (1959, p. 31). As the main objective is to understand and distinguish mathematics cultures (secondary and tertiary cultures) through teachers' WDMs, working with teachers from both levels would create an interesting contrast that could bring the specific WDMs to light. As will be explained,

Table 1 The concept of mathematical ethnomethods (Corriveau, 2013)

Ways of doing mathematics:	What teachers actualize:
Reflexive and accountable (describable, intelligible)	Practical reasoning (interpretative procedures) and rationality, inseparable from the action/which give meaning to and create the action
Indexical	Ability of teachers (as members) to resolve indexical exchanges relating to the action and to recognize the circumstances of the action
Shared	Competence of teachers as members

this choice is also relevant from an ethnomethodological perspective. In fact, it leads to what we call breaching situations (Garfinkel, 1963).[4]

4.1 A collaborative research

The model of collaborative research (Bednarz, 2013; Desgagné, Bednarz, Couture, Poirier, & Lebuis, 2001), in the tradition of interpretive research, is "aimed at bringing the world of research and the world of practice closer together and mediat[ing] between these two cultures, in order to construct knowledge leading to informed practice" (Savoie-Zajc & Bednarz, p. 578). The *reflexive activity*, in which researchers and teachers interact over an aspect of practice, is the central idea within this collaborative research.

4.1.1 The reflexive activity

The reflexive activity is rooted in ethnomethodology and relies on the concepts of reflexivity and accountability (Bednarz, 2013). Reflexivity and accountability are associated with action. The reflexivity means that the action accounts for and, at the same time, creates the world. This means, from a methodological point of view, it is necessary to put teachers in an action situation so they *account for* their WDMs while establishing them. The challenge here is to develop a reflexive activity using situations within the limits of what is familiar to the participants in their teaching practice (Desgagné, 2001).

In this regard, I used situations that teachers encounter daily in their work. For example, I developed different situations in which teachers had to make sense of a mathematical problem or question in a teaching context, comment on it, and explain how they would carry out a task in their classroom. They would anticipate students' solutions to a problem, elaborate on excerpts from textbooks, comment on students' productions, explain a part of their lesson, etc. The idea was to identify, through these different accounts of actions, how teachers do mathematics.

Also, since teachers from one level are not necessarily familiar with the other level, putting teachers from both levels together encourages breaching situations. The circumstances created by reactions of teachers from one level to unusual situations at the other level can reveal the meaning they attach to their ways of doing mathematics in class, enabling teachers to explain assumed facts and shared ways of doing. In class, teachers do what they do naturally and without conscious thought; confronting teachers with unusual situations forces them to explain the meaning behind their habitual and familiar actions. Situations were built around this familiar/unfamiliar fine line (see § 3.1.2).

4.1.2 The specific investigation

This study takes place in a specific context, in Quebec (Canada), where there is an obligatory 2 years (years 12 and 13) of preparatory college for students planning to attend university.[5]

[4] See Corriveau, 2013 for further information about the use of breach or Herbst and Chazan (2015) who have also use the concept of breaching experiment (from ethnomethodology) in their work.

[5] It may seem difficult to talk about the phenomenon of inter-level transition from an international point of view, since educational institutions, as Gueudet (2008) points out, differ from one country to another.

This pre-university program follows secondary level (grades 7 to 11). It is attached, as well as the university, to tertiary (higher) education. The college institutions, created in the late 1960s, are independent from the secondary institutions and from university and lead to a specific diploma (collegial studies diploma). College teachers have a disciplinary training (usually a master's degree in mathematics, literature, history, geography, etc.) and have access to research grants. But unlike teachers at the university level, they are not required to do research.

A group of seven participants (four secondary teachers and three tertiary teachers) were invited to join a collaborative research project, in which the researcher worked with practitioners on an issue related to their practice (Desgagné et al., 2001). Six teachers participated until the end of the research project. The project ran from November 2010 to November 2011. There were regular meetings: one meeting to explain the project and 6 days (9 a.m. to 4 p.m.) of conversational meetings. Different topics were addressed during these meetings: functions, the use of symbolism (see Corriveau & Bednarz, to be published) and proofs. All these meetings were recorded and transcribed.

Once teachers had agreed to participate, the group was formed, and a first meeting was scheduled to explain the project, the first situations used as a basis for discussion had to be thought of. Below, I present an example of a situation given to teachers during the first meeting. This is a situation with two familiar tasks insofar as the tasks relate to secondary and college content (functions); they refer to usual representations of function used by these teachers with their students at the secondary level (table of values and graph) and at the college level (table of variation and graphics). It is also a breaching situation because Task 1 is likely to be done at the secondary level and Task 2 introduces a representation only seen in college. It is also a breaching situation because, although these tasks could be given to school students, no task of this type was found in the textbooks of either level or carried out in class during observations.[6] The fact that the teachers could be destabilized by the tasks allowed for the explicitation of usual WDMs (as seen in § 4.2).

Example of tasks given to teachers as a base for discussion For example, as a starting point to open a discussion, I presented the following tasks taken from Coppé, Dorier, & Yavuz (2007)[7]:

Task 1 The following table represents a function f which is defined on [−3, 3].

X	−3	−2	−1	0	1	2	3
$f(x)$	2	1	−1	0	0,5	1	2

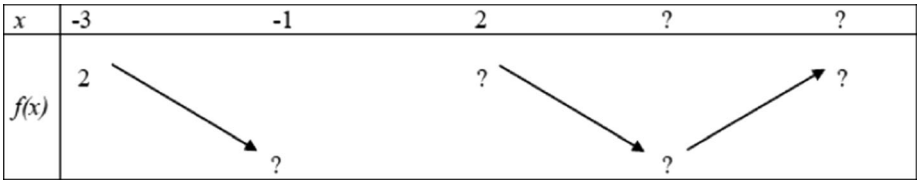
a) Draw a graph compatible with this table.
b) Can we draw other graphs? If we can, draw one. If not, explain why.

[6] To design tasks as a basis for discussion, I observed mathematics classes at both levels.
[7] This breaching task could also be used with students entering the tertiary level. In fact, as suggested and used by Praslon (2000), unusual tasks make students sensitive to the gap between the secondary and the tertiary levels.

Task 2 The following table represents a function *f* which is defined on [−3, 3].

X	−3	−2	−1	0	1	2	3
$f(x)$	2	1	−1	0	0,5	1	2

a) Complete the table of variation so it is compatible with this table.



b) Are there other ways to complete it? If yes, which ones. If not, explain why.

Then, I asked the following questions (participants worked in small groups of secondary/college level teachers):

Questions on the task
- Is it plausible to find a task like this at the secondary level? At the college level?
- What could be the purpose of this task?
- How could you use this task at the secondary/tertiary level?
- I also provided students' answers to facilitate further discussion (Appendix A).

Questions on student answers
- What would be a possible answer given by your students?
- Based on the student work provided, what conceptions can be induced about functions by these tables?

4.2 Research material and analysis

The main material used in analysis was a transcript of the reflexive activity.

4.2.1 Preparation of the analysis

Before beginning the codification process, all the meeting recordings had to be viewed and the audio transcribed.

- *Preparing the transcripts and a first selection of relevant extracts*: I watched all the video recordings, deleting all discussions not directly related to research. Then, during the course of the viewings, major exchanges caught my attention. These were general excerpts in which teachers talked about mathematical content and processes. When such moments were targeted, I kept the entire segment. This first step was very useful in identifying what

is shared among teachers of a specific level (a central dimension for the purpose of the research). Familiarity was revealed not only through words but also through gestures, expressions, eye contact, etc. Thus, this repeated viewing not only helped me to become familiar with the data, but also to retrace different indicators to support this dimension of sharing (gestural and verbal confirmation were noted in the transcription).

- *Presentation of the material in the transcripts*: For purposes of presentation in this paper, the secondary teachers' dialogue and that of college teachers have different codes (color codes). While I was carrying out the coding and analysis, it appeared important to unify the color to really feel that the discussions were the result of several "voices." The color code therefore appeared later in the process, making it easier to identify the different voices (teachers from secondary or collegial level, researcher). Fictitious names were used. Secondary teachers have their first names starting with the letter S (Scott, Sergio, Sam, and Sandra). Those of the college level have a name beginning with Co (Colett, Colin, and Corinna).
- *Identification of themes in the transcripts*: An analysis was already going on during the elaboration of the meetings and during the evolution of the sessions (placed in the research journal of the researcher at the end of each session). Thus, during the first meeting, themes were identified (the work with functions, the use of contexts, demonstration, symbolism) for further exploration in subsequent sessions. After listening to the recordings and in the light of the field notes, several themes were selected for analysis, one of which having occupied much space in these meetings: the work around the concept of functions and its representations.

4.2.2 Analysis approach

The analysis focuses on different representations of functions (the table of values, rate of change, graph, and symbolic expression) and clarifies what using these representations to do mathematics means. I gathered transcripts from sessions during which functions were discussed and coded all such incidents. The ethnomethodological conceptualization that emerged from the coding led us to re-establish several theoretical concepts that were useful to consider during analysis (WDMs, interpretative procedures, indexicality, rationality, etc.).

The data collected was analyzed at several levels. Primary analysis was guided by the question, "What do teachers of a particular level impart about their ways of doing…?" This initial analysis represented the voice of actors "as delivered" (Demazière & Dubar, 1997; Desgagné, 2007). During the subsequent stage of analysis, I attempted to understand the results of the primary analysis of both levels on the shared WDMs demonstrated by teachers involved in this study. Next, I looked at the data in terms of the transition (differences between levels), partially through a concurrent interpretation of results from both levels, as well as through entrance into the culture (as defined by Hall). At this second stage of analysis, I adopted an analytical stance (Desgagné, 2007), in which the researcher proposes an interpretation of the primary analysis in order to further advance the understanding of the transition.

5 Analysis of mathematical ethnomethods actualized by teachers, specifically relating to functions

An initial review of the transcription of the first meeting shows WDMs that, in the teaching context, can be linked to the institutional aspects of work with functions as they are explained

Table 2 Excerpts from secondary and tertiary curricula (MELS, 2010; MELS, 2012, Quebec, Canada)

Secondary cycle two curriculum	Tertiary curriculum
In secondary cycle two, students hone their ability to evoke a situation by drawing on several registers of representation and switching from one register to another, without any restrictions. For example, functions may be represented using graphs, tables or rules, and each of these representations conveys a specific point of view and is complementary or equivalent to other representations. Students learn to analyze and deal with situations that involve a set of algebraic concepts and processes. They establish dependency relationships between variables; model, compare and optimize situations, if necessary; and make informed decisions about these situations, depending on the case (MELS, 2012).	Elements of the competency (recognize and describe the characteristics of a function expressed in symbolic or graphic form). 1. To recognize and describe the characteristics of a function expressed in symbolic or graphic form. 2. To determine whether a function has a limit or is continuous or has a derivative at a point or on an interval. 3. To apply the rules and techniques of differentiation. 4. To use derivative and related concepts to analyze the variations of a function and then graph it. 5. To solve optimization and rate of change problems (MELS, 2010).

in secondary and tertiary curricula[8] (see Table 2). It is therefore appropriate to review these aspects in order to demonstrate how WDMs are enacted, before introducing the implicit element of mathematical ethnomethods.

There is no doubt that secondary and tertiary teachers are not engaged in the same ways of doing mathematics in respect to institutional expectations. The following section shows the overall results of the analysis, first at the secondary and then at the tertiary level, excerpted from the session transcript.[9]

5.1 Secondary level

It is no surprise, according to institutional expectations, that secondary WDMs are organized around the functional modeling of diverse situations systematically leading to functions belonging to some few families approached through tables, graphs, and algebraic expressions. The WDMs are illustrated with excerpts taken from discussions.

– To connect a table of values, graph, or symbolic expression to one of the function models provided (basic functions).

Sam	[Looking at the table of values, cf. Task 1, § 3.1.2] I'd be curious to see what this gives to start with [the function we're working with is implied]. [Later, in an account where he describes what he does with a table of values] In Year Three, when […] we give tables of values, we ask […] if it's a linear function, try to find which function it relates to based on the table of values.
Sandra	[Looking at the table of values, cf. Task 1, § 3.1.2] Okay, it goes down to here… I thought it could be an absolute value, but it's not.

[8] It is not the case for other themes analyzed in the larger research project, i.e., the use of symbolism and the use of contexts in mathematics. These transversal themes are not as well detailed (as a function, as content) in curricula.

[9] The sessions were carried out in French; the excerpts which follow are translations.

– To sketch (the general appearance of) a graph based on a table of values or symbolic expression (using the parameters).

Scott	Based on a table of values, or from the context, we could try to see what rate of change it is and see how the value of the function changes when x changes in a particular way.
Sergio	[Referring to the graph] That's it. You change different parameters (of a base function) and you can see that there are some things that still… Rate of change characteristics don't change.
Sergio	…then, we establish that $f(x)$ equals x squared [$f(x) = x^2$]. Then we can… Sometimes, we can get there by observing characteristics, like graph characteristics, two symmetrical branches, a vertex. All those but also rate of change characteristics.

– To establish links between various representations (tables of values, graphs, and symbolic expressions) and function characteristics (rates of change, vertices, asymptotes, etc.)

These different ways of doing are linked to a specific rationale that gives them meaning. Secondary teachers have expectations for students that are strongly expressed in the recognition of "families of functions," as shown below by Scott and Sandra.

Scott	At the secondary level, I'm happy if students finishing high school can, based on a situation, a table of values, even an equation, […] make a table of values and recognize the type of function; if they can recognize that this one is rational, that one is exponential based on the rule. They should have a solid understanding of this, at the very least.
Sandra	When we study a function in secondary school, (a particular function), it's normal for them to have trouble when… [addressing Corinna] you mix a bunch of functions together, basically, right?

5.2 Tertiary level

At the tertiary level, the approach of functions is typically a Calculus approach, with an emphasis on local properties and the power given by the notion of derivative.

– To describe the behavior of any function based on a symbolic expression or graph.

Corinna	[Based on a graph] [E]xactly. When it increases faster and faster, the growth rate, how we could calculate it… I work with slopes, a lot of drawing slopes […]

– To utilize tools to describe this behavior and sketch its graph. In the following excerpt, Colett shows that the rate of change is one of many conceptual tools, such as derivatives, which can be used as a guide to sketch the graph of a function.

Corinna	There's a purpose to doing the rate of change [cf. Task 1, § 3.1.2]. Starting with the study of signs, derivatives, second derivatives, they have to find the domain first, they have to sketch the graph of a complex function. It's not just a quadratic function, there are all sorts of elements mixed in there [addresses a secondary-level teacher] and… the only way to figure it out is the rate of change. And on top of that there's concavity; arrows aren't linear functions, there's concavity.
Colin	That's right. For (students), what we do with the rate of change makes sense.

– To predict the *limit behavior* of a function at any point, based on various representations.

Colin	[Regarding WDMs with functions at the secondary level] We don't look at the rate of change between each value (like you do at the secondary level with the table of values), we look at where it's going to end up. It's a bit different…
Corinna	[Based on the graph] I work with slopes, a lot of drawing slopes; it's the same growth rate [refers to two curves]. If we look at a specific moment, then oops! We're going to get something else. So working with that…

As at the secondary level, these tertiary WDMs overlap with and are shaped by a specific rationale of institutional requirements: concepts (derivatives, limits, etc.) and representations (tables of variation) are envisioned as tools employed to accomplish the objectives of differential and integral calculus courses:

Corinna	At the tertiary level, the goal of the first calculus course is to use the tools to be able to… One of the goals is to sketch the graph of a function, to recognize the behavior of the function at almost any point. […] If I ask them to sketch the square root of x [$\sqrt{x}$], they'll be able to. But if I make it more complicated, like the root of x^2 + something [$\sqrt{x2 + \ldots}$], then I don't know what it'll look like. Based on an algebraic expression, they'll have to figure out for what values this function is defined. Are there discontinuities? Are there asymptotes? Increases, decreases? On what interval?
Colett	Whatever function we give them, created based on known functions… At the tertiary level, we have derivatives, a tool that lets us undertake any function, while at the secondary level you don't. That's why you deal with certain types of functions.

This initial identification of WDMs at each particular level for functions and representations highlights certain distinctive characteristics that allow us to distinguish between the two. Teachers are enacting WDMs that remain similar to what could have been concluded from an institutional analysis. Data on the breaching situation (see § 3.1.2) allows to refine the analysis and bring to light subtle distinctions between the two levels. The analysis is done by means of indexicality, in terms of interpretive procedures.

5.3 Analysis of interpretive procedures for entering the realm of the implicit

Analysis of teachers' interpretive procedures clarifies other WDMs and also makes it possible to observe *ways of investigating* which overlap with WDMs. In the following pages, I will revisit ways of investigating employed by teachers at both levels in respect to several topics relating to functions and representations. In doing so, we shall enter into the realm of the implicit in order to better understand transitional issues.

5.3.1 Ways of investigating that give meaning to the table of values

Secondary and tertiary teachers immediately start with an interpretation of the table of values (Task 1, § 3.1.2), in order to give it a meaning that corresponds to their WDMs. In the table below, quotations are placed in side-by-side to illustrate the different ways teachers at the two levels indexed meaning in the breaching situation. Table 3.

For secondary teachers, the table of values represents the function being taught, so they use their ways of approaching and investigating the table of values to look for a

Table 3 Excerpts from transcripts illustrating the ways of giving meaning to the table of values

Secondary	Tertiary
I'd have to see what kind of function it is. For sure it's a modeling problem. For sure when it's a modeling problem, you're supposed to see a function [in reference to the function being taught]. (Scott)	What I find interesting is that at the beginning, it's just a bunch of pairs [in reference to points] that we place and can link in many ways, in different ways. (Corinna)
Okay, it goes down to here [indicates specific values in the table]. I thought it could be an absolute value, but it's not. (Sandra)	Working with the table of values to represent functions doesn't interest me, because it's not comprehensive enough, but I understand that students have to get to the point where they understand that a function isn't defined by specific points only. They have to be able to understand that it's only part of the function, and that there's lots going on in the rest of the function. Once they understand that, the table of values is just some points of the function… (Colett)
I'd be curious to see what happens when we start with the kind of function we're usually working with. (Sam)	

model of the function it represents. This way of investigating demonstrates a WDM (associate the table of values with a function). The table of values, therefore, is indexed to a particular meaning for secondary teachers, the representation of one (and only one) function. For tertiary teachers, the table of values is a collection of points "This is not the function," so their ways of approaching and investigating it are quite different. In a way, they see everything that is not there ("[It's] some points that can be connected in different ways"; "A lot of things can happen"). This way of investigating also demonstrates a WDM (use a tool, in this case the table of values, to describe the behavior of a function). The table of values is therefore indexed to a different meaning than at the secondary level. It becomes a tool for gathering information, in this case, points appearing on the graph of a function. Interpretive procedures emerge at this stage, integral to mathematical ethnomethods and overlapping with ways of doing (Table 4).

This particular way of investigating the table of values leads teachers at both levels to reject the task as proposed; it is unfamiliar for both secondary and tertiary teachers. Teachers from both levels explain the reasons for rejection, and in doing so, the circumstances outline the frontiers of the mathematical ethnomethods on the table of values. Secondary teachers indicate their rationale for rejection to be a lack of correct circumstances (insufficient information in the

Table 4 Interpretive procedures associated with WDMs with the table of values, for secondary and tertiary teachers

Interpretive procedures integral to mathematical ethnomethods	Secondary teachers	Tertiary teachers
Way of giving meaning (indexicality) to a table of values	Represents the function	Is a tool for gathering points on the graph of a function
Way of investigating the table of values	Teachers look for the function represented by the table of values.	Teachers see points that are and are not shown on the table of values
Overlap with a WDM with representations	Connect the table of values to a function	Use tools to describe the behavior of a function

table of values to recognize a model; no context for validating the model), leaving them unable to apply their familiar WDMs. However, they explain that the table would be acceptable if the table of values represented a function being taught. Tertiary teachers, in rejecting the task, also explain circumstances in which the table of values would be acceptable to use: "We use the table of values to evaluate a limit, to calculate the value of images, when we have a function that is not defined for a particular value," Corinna explains. "What happens if we approach a certain value?" In this case, reading the table of values is part of a process. The table of values, in addition to representing the points on the graph of a function, represents the process of closure.

5.3.2 Ways of investigating the domain of a function

Teachers at the two levels indexed the following information differently: "Let *f* be a function defined on the interval of −3 to 3 […]" (See Table 5).

These discussions reveal the meaning assigned to a domain as inseparable from ways of doing previously demonstrated by teachers. When questioned about the domain, secondary teachers indexed it to a specific meaning, the set upon which a function is defined. Their way of investigating a domain focused on whether or not it was defined by real or integer numbers; on whether or not the points should be connected. The interval corresponds to the known part for which information exists. It suggests that the function exists "before −3 and after 3" but we do not know its behavior. For tertiary teachers, the domain is the interval upon which the function is defined. A way of investigating a domain is based on continuity as Colett states: "Because it's defined as continuous on the interval" and she also adds later "I don't know if [the function] is continuous." Their way of giving meaning to a domain is clear from the way they read student answers (Appendix A). For example, Colin says, "That's funny, many of them exceeded −3 to 3. For us [tertiary teachers], it's not really good for them to exceed the −3, 3 interval […] in optimization problems and all that."

Table 5 Excerpts from transcripts illustrating ways of investigating the domain

Secondary	Tertiary
Does "defined on the interval of −3 to 3" mean that it's a linear function or just points? (Do we consider them real numbers or integers?) (Sam)	Because it's defined as continuous on the whole interval. What I understand by that is that they have to continue something based on (the table of values). (Colett)
If the function is defined on IR, it starts at −3 and goes to 3. (Sandra)	When I see "defined from −3 to 3," to me that means that the domain is the interval. (Colett)
[O]f course we say it's defined "from −3 to 3," but there's nothing that says if the function is defined based on real numbers or integers. (Scott)	In CEGEP [college], we define on the interval [referring to the domain— This answer was in response to a question from the researcher about ways of discussing the domain]. (Colin)
We aren't necessarily saying that it's the domain. In less mathematical language I would say that before −3 and after 3 we don't know, we can't tell. (Scott)	

Table 6 Excerpts from the transcripts illustrating ways of looking at graphs

Secondary	Tertiary
Actually, almost all the answers between the first two points are linear functions. Is that really it? Why would it be that instead of something else? (Scott) What I found interesting is that when we sketched it, I asked myself is it only a line? What's between two points? You [the searcher] came to us with your examples (of student answers)… That would be workable sometime in a class. But like that? (Sam)	No, I don't think it's continuous. It's just saying that at each point, there's a value… and here are a few. So there are some holes to fill in different ways […] Then all the students did continuous functions. (Colett)

5.3.3 Graphing at both levels

Teachers had different approaches to graphing (see Table 6).

The secondary teachers had a specific approach to sketching of a graph from a table of values. They focused on what connects, or does not connect, the points. This is evident in the way teachers read the student answers, a tendency observed in linearization. Scott noted that students "connect the points with line segments, like a line graph." This approach can also be observed in their questions on the type of function between two points. Sam stated, "Between two points, we can draw several types of functions. Interesting!" This way of approaching the graph to be drawn from a table of values clarifies WDMs identified previously (sketch a graph from a table of values). Excerpts from other tasks improve the analysis of ways of looking at graphs. For example, teachers add function-specific (graphic) characteristics into a graph or symbolic expression to teach students to recognize the appearance and model of the function from the vertex of the quadratic function (and its two symmetrical branches), the two pairs of asymptotes in the rational function, a rate of change, and the effect of parameters. When I asked how we can tell that a function belongs to a particular family, Sergio said, "By observing its characteristics; it has graphic characteristics, two symmetrical branches, a vertex [in the case of a quadratic function, for example]." Sam added, "When a dozen textbooks say that the vertex of a quadratic function is (h, k)." These ways of investigating are linked to ways of doing: sketch a graph based on a function written symbolically and link representations through characteristics of a function (rate of change, vertex, appearance, etc.)

Tertiary teachers also approach sketching graphs from a table of values in a particular way, focusing on the continuity, or discontinuity, of the function. This is evident in their reactions to student answers—they were surprised that students produced continuous functions—and their explanations. For example, Colett said, "No, I don't think that it's continuous, it's just saying that at each point there's a value, and here are a few [points]. So there are some holes to fill in different ways." She explained what "defined on the interval −3 to 3" means, to reveal, or demonstrate, the possibility of discontinuity. She was not concerned with the different ways of connecting the points from the table as much with knowing if the function would be continuous on the subintervals between the

points. This tertiary way of investigating also explains previously identified WDMs (predict the limit behavior of a function at any point based on different representations).

One approach highlights the possibility of connecting the points in different ways, using the concepts of rate of change and models of known functions that they try to find (how it changes between two points). The other approach introduces the possibility of discontinuity, asking about a possible discontinuity in line with what it means to be defined on the interval.

6 Interpretation and discussion

The previously demonstrated ways of doing and ways of investigating (mathematical ethnomethods) are established based on distinct cultures at each level. I will now conduct a concurrent examination of these mathematical ethnomethods in order to further the transition issue.

6.1 The cultural perspective

The previous analysis provides a glimpse into two cultures with different WDMs with function and different ways of investigating and giving meaning to their representations. The technical dimension of culture (the explicit and the institutionalized), especially notable in teachers' rationales, acts as a structural resource (Lave, 1988) for the informal dimension, i.e., the WDMs). However, the technical dimension is guided by a series of small details from the informal level. The analysis therefore shows that teachers maintain flexibility and subtle ways of investigating and give meaning to objects implied by the specific context and circumstances of secondary or tertiary education. Each level's mathematics culture in relation to functions, representations, and associated concepts is addressed from the perspective of this overlap of technical and informal cultures. How do these cultures distinguish and structure themselves at each level?

- ***Same representations***, ***different meanings***: At the secondary level, a representation portrays in some way a function recognizable through characteristics specific to this representation. For example, the vertex of a quadratic function is visible on the graph, of course, but also in the symbolic expression (teachers use canonical form). Teachers recognize the sketch of a graph and the effect of its parameters in the symbolic expression; they recognize the rate of change by the appearance of a curve (visually), and in successive numerical bounds between values in the table of values (numerically). For tertiary teachers, distinctions must be established between representations. Contrary to at the secondary level, each method is not the "faithful" representation of a function. Implicitly, preferred representations are those that can be adapted to their ways of doing (describe the behavior of any function and predict the limit behavior in detail). Some representations are used as tools, such as the table of values and the table of variation. For example, the table of value at this level is

mainly used to represent a process of closure. It allows an understanding of what is happening at a very small scale. The symbolic expression and its graphic representation are "faithful" representations of a function.

- ***A domain known by the type of function*** **versus** ***a domain given by an interval***: Teachers investigate domains differently. At the secondary level, once teachers have found the function represented by a table of values or by its algebraic expression, they also know the domain. They only have to determine, from the context, if it is defined on real or integer numbers. For college teachers, the domain is given by the interval and they wonder what the behavior of the function is. Investigating the domain is part of investigating the function.
- ***Portraying families of functions*** **versus** ***changing scales***: The portrait is a relevant metaphor for visualizing the mathematics culture surrounding functions at the secondary level. Teachers paint a mental picture of each function taught, in which each representation presents a particular angle of the whole picture. We can also portray the entire family of functions through each representation. The goal, for secondary teachers, is to identify particular families of functions through these different portraits (table of values, graph, symbolic expression, or even a through a context). The concept of the portrait also makes sense when the "focal length" of the image is fixed. The overall characteristics of each function are of interest. Secondary teachers do not need to change the focal length (contrary to tertiary teachers). It is not necessary to investigate a function in fine detail, since the behavior, domain, vertices (if there are any), and asymptotes (e.g., associated to rational function) of the function are already known. Everything is known, so all that is left is to identify the family through the various representations. Zooming in and out is a useful way to describe the tertiary culture surrounding functions. A function is investigated by means of its symbolic expression or graph using different tools, but also at different scales. Although it seems obvious that sometimes teachers attempt to reveal the overall appearance of a function and try to determine its local behavior, it has implicit impacts on the meaning given to representations.

  When the graph of any function is presented (or when students are asked to create one) the possibility of discontinuity, rate of change changes, etc. must be explicitly noted. In the same way that a triangle must be generic when used as the basis for the reasoning in a demonstration, the graph of any function at the tertiary level must be generic (as has been observed in the reactions of tertiary teachers to student answers). It is a new way, for students going from secondary to tertiary level, to consider the graphic of a function. Going from secondary to tertiary level leads students to new ways to investigate representations.

6.2 Including teachers in the analysis of transitional issues

This analysis does not radically change what has already been said about transitional issues and working with functions, but allows to highlight new differences between the secondary

and tertiary levels. For example, the analysis made by Vandebrouck (2011a) using the definition of Robert's "field of work"[10] (2003) allows the author to specify three large fields related to work with functions in the school system. Vanderbrouck was able to constitute three fields from an analysis of the French curricula, textbooks, and courses notes. The first two fields, called F1 and F2, are quite close to what we could find from an analysis of official documents in Quebec (at secondary and college level). Vandebrouck explains that the first domain (F1) is the reaching of a first level of conceptualization of the notion of function. Many concepts such as periodicity, growth, minimums, maximums, domain, co-domain, and asymptotes are introduced in many types of representations (graph, table of values, symbolic expression). One could add that these concepts are introduced while familiarizing students with basic functions (linear, quadratic, rational, exponential, logarithmic, trigonometric). In this field, the conceptualization of function is done through specific functions. Vandebrouck explicates that the second field (F2) unifies and simplifies field F1. Concepts such as limit, continuity, and derivability are introduced in this field. The work is more algebraic than the work done in field F1. Referring to Coppé et al. (2007), Vanderbrouck explains there is a strong algebraization of techniques, based on rules and algebraic calculation (evaluation of limits, derivatives, study of variations, polynomials, exponential, logarithmic, etc.).

There are already a lot of differences between the two fields but, technically, in a cultural sense, it seems obvious that college is in continuity with the secondary level since F2 unifies and simplifies F1. However, although the technical dimension acts as a structuring resource and somehow structures the culture constituted at each level, a series of small details in the informal dimension of these cultures shows leeway of teachers, and that this culture is organized also in the implicit. Our study demonstrates that even if the same concepts are used at both levels, e.g., the concept of "domain," they are conceptualized differently. We may also put forward the hypothesis that the same is true for other notions, such as the growth of a function, its positivity, continuity, etc. This reading also highlights that the significance of representations, for those who use it, ties into the context in which they are used. Teachers from the two levels both use representations of functions, but the ways to investigate these representations, the way teachers use it, lead to interesting distinctions.

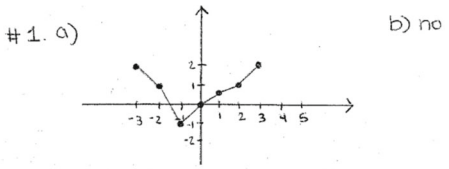
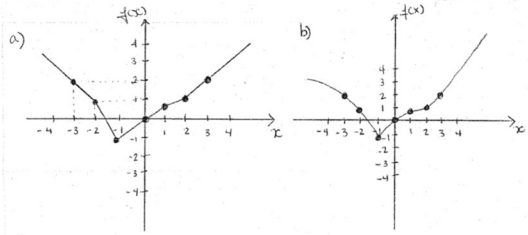
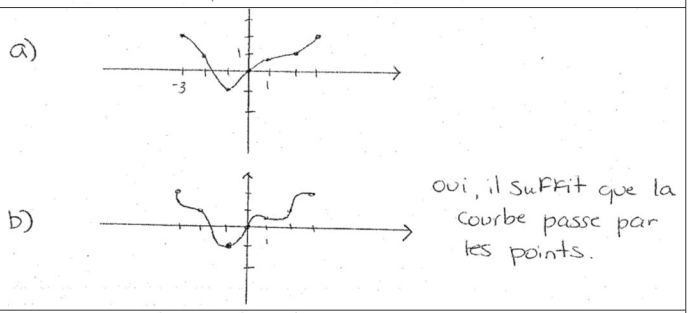
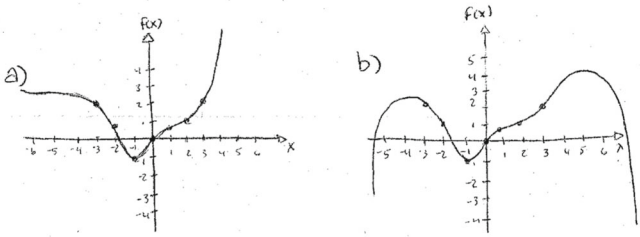
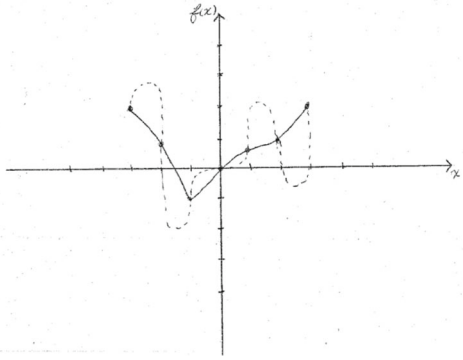
7 Conclusion

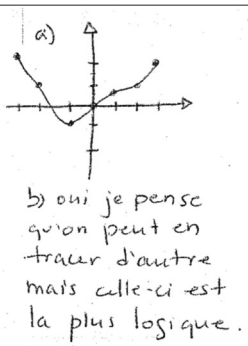
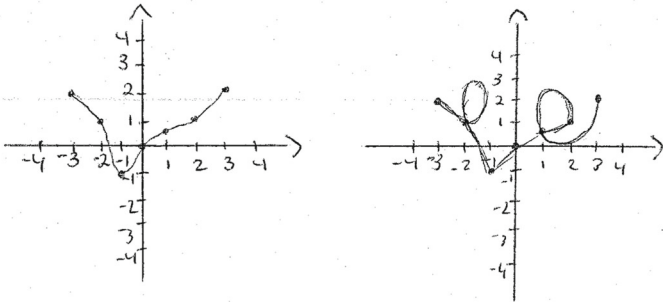
At the beginning of the article, I pointed out the importance of including teachers in the analysis of transitional issues. This involvement of teachers leads to an understanding of what students must face when entering the tertiary level. Indeed, the significance of representations and associated concepts ties into the context in which they are used. Teachers might use the same representations (e.g., the graph) when working with functions, but the circumstances in which these representations are used, how teachers use it and how they investigate a function with it, leads to important distinctions. One might think a priori that concepts like domain, continuity, change of rates, etc. are invariant regardless of school level or context. The analysis conducted here shows that, rather than depending on the ways teachers use these concepts (deeply entrenched in institutional considerations), the meaning changes slightly in the passage from one level to the other.

[10] Translation of the French term "*domaine de travail*."

Appendix A

Students' solutions

Student	
A	
B	
C	
D	
E	

F	
G	

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